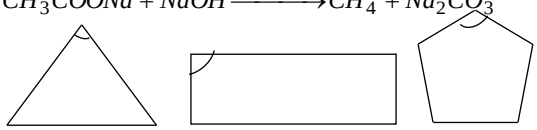
141. (a) Difference in the rates of diffusion of gases — The marsh gas (methane) detector works on Graham's law, based on different diffusion rates of gases.

142. (b) $CH_3COONa + NaOH \xrightarrow{CaO} CH_4 + Na_2CO_3$

143. (a)



Cyclopropane with a bond angle of 60° is very strained and hence very reactive.

144. (c) CH_4 — Methane requires UV light for free-radical substitution; it does not react with chlorine in the dark.

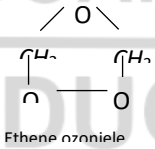
145. (b) Marsh gas mainly consists of methane.

146. (c) $CH_3 - Cl + 2H \xrightarrow{Zn/HCl} CH_4 + HCl$

147. (a) Methane — Methane is the most stable alkane because it has the lowest heat of combustion per carbon and no C – C bonds, making it least reactive.

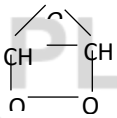
149. (d) Propane is a saturated compound. Ozonide is formed only by alkenes or alkynes

$CH_2 = CH_2 + O_3 \rightarrow$



Ethene ozonide

$CH \equiv CH + O_3 \rightarrow$



Ethyne ozonide

150. (b) $H - \overset{sp^2}{\underset{\underset{H}{|}}{C}} = \overset{sp^2}{\underset{\underset{H}{|}}{C}} - H$ sp^2-hybridisation (trigonal planar).

151. (a) $Ag - C \equiv C - Ag + 2HCl \rightarrow CH \equiv CH + 2AgCl$

152. (c) Wurtz reaction

$$C_2H_5 - I + 2Na + I - C_2H_5 \xrightarrow[\text{Ether}]{\text{Dry}} C_2H_5 - \underset{\text{Butane}}{C_2H_5} + 2NaI$$

153. (a) Molecular solid — Solid methane consists of discrete CH_4 molecules held together by weak van der Waals forces, so it is a molecular solid.

154. (b) All the $C-C$ bond are single bonds. Hence sp^3 -hybridization and tetrahedral structure.

155. (a.b) $CH_3MgI + CH_3-CH_2-NH_2 \rightarrow CH_4 + CH_3CH_2NHMgI$

$CH_3MgI + C_2H_5OH \rightarrow CH_4 + C_2H_5OMgI$

Alkyl group of Grignard's reagent is involved in the formation of alkane.

156. **(c)** **B_2H_6 / NaOH, H_2O_2** — Hydroboration – oxidation of propene gives anti-Markovnikov product, propane-1-ol.

157. **(b)** **Cracking** — Controlled pyrolysis breaks higher hydrocarbons into lower hydrocarbons.

158. **(a)** **$-CH_2-$** — Successive members of alkane homologous series differ by a $-CH_2-$ group.

159. (a) General formula of alkane C_nH_{2n+2} (n = no. of atoms).

160. (c) $CH_3Br + H_2 \xrightarrow{LiAlH_4} CH_4$ (methane) $\xrightarrow{Na} CH_3-CH_3$ (Ethane)

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